Lesson 05 Demo 01

Implementing a React Application Using the useEffect Hook

Objective: To build a React application using Vite and implement the useEffect Hook to fetch and display data dynamically, ensuring efficient state updates upon the component mount

Tools required: Visual Studio Code and Node.js

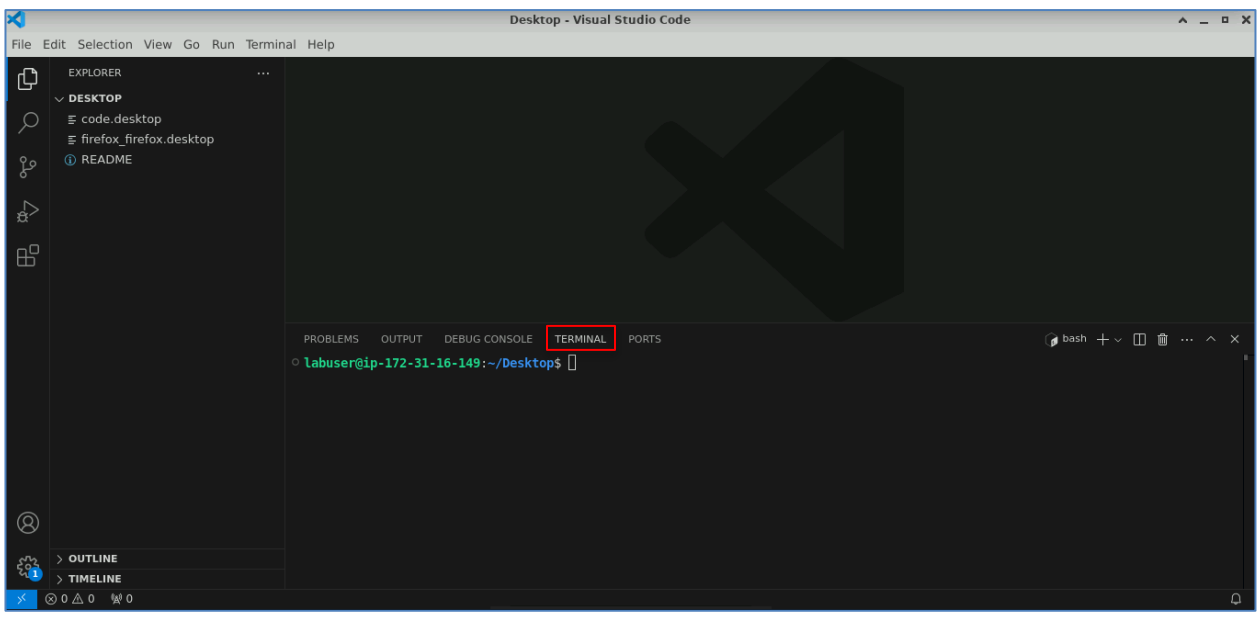
Prerequisites: None

Steps to be followed:

1. Set up a new React project using Vite
2. Modify the App.jsx file
3. Run and verify the application

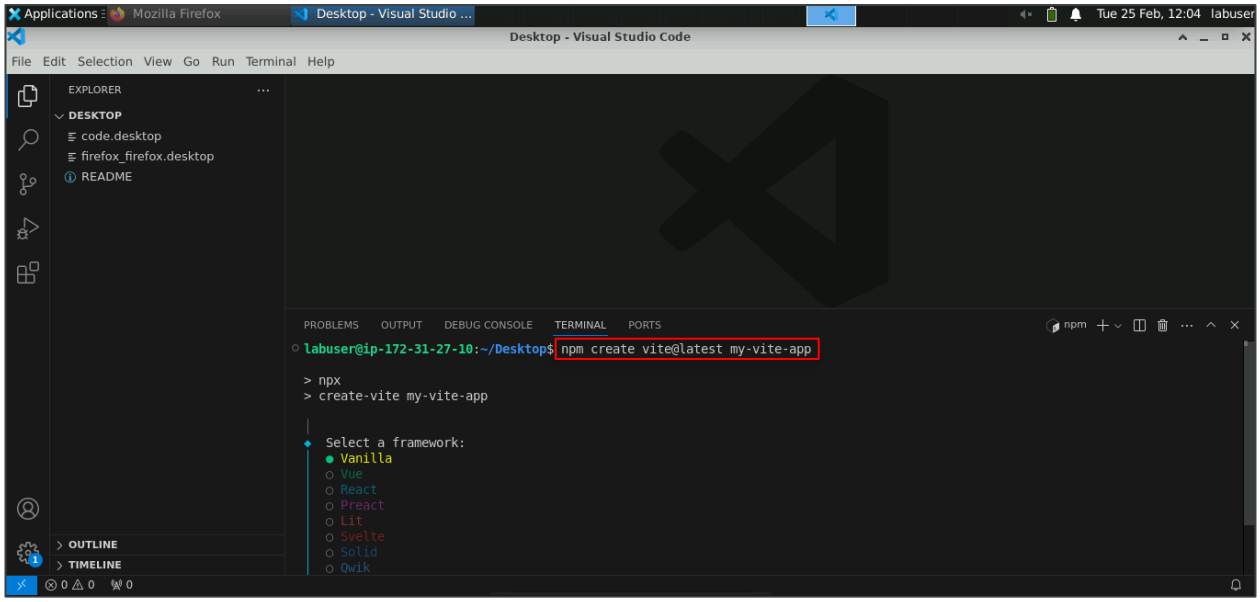
Step 1: Set up a new React project using Vite

1.1 Open Visual Studio Code (VS Code) and launch the terminal

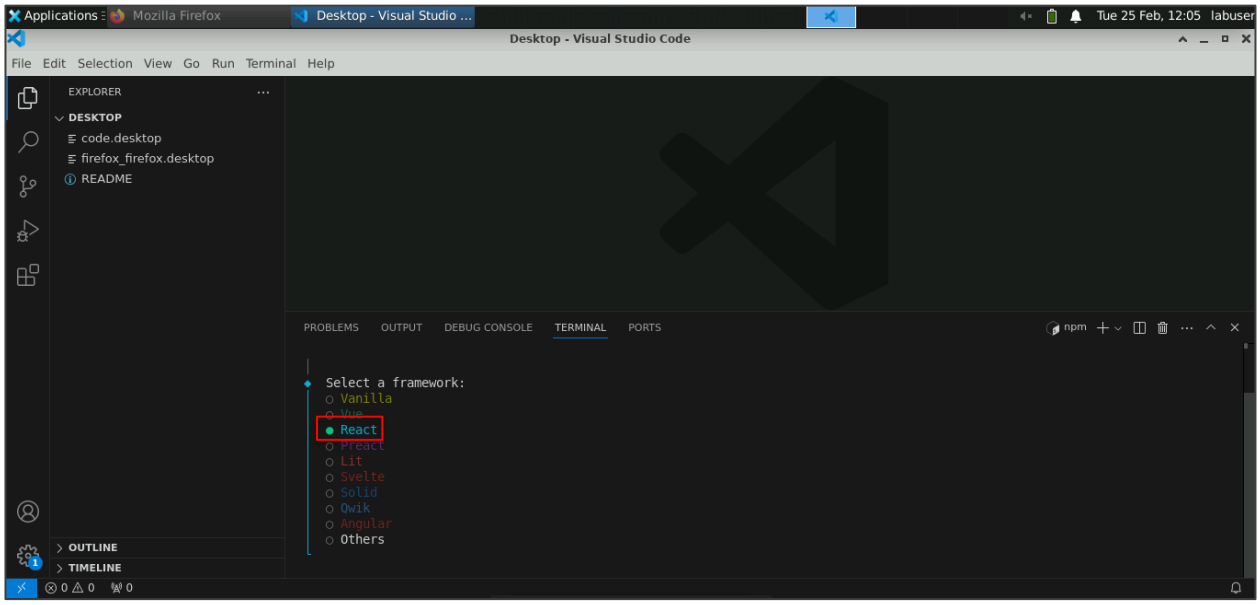


1.2 Run the following command to create a new React project using Vite:

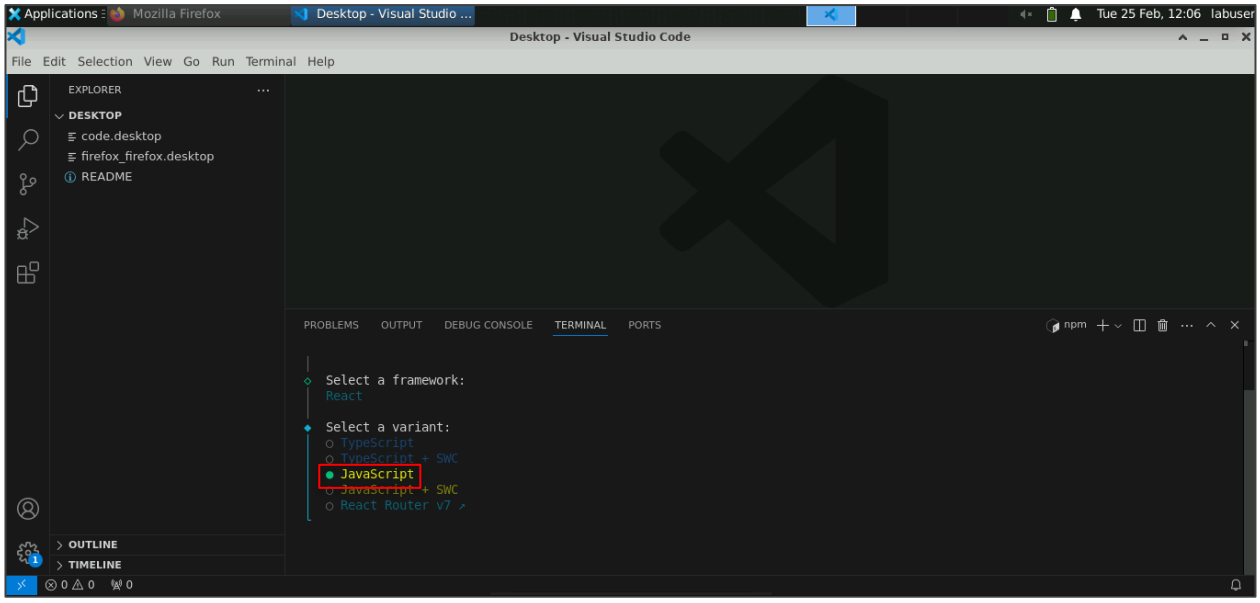
npm create vite@latest my-vite-app



1.3 Select **React** as the project framework and press enter

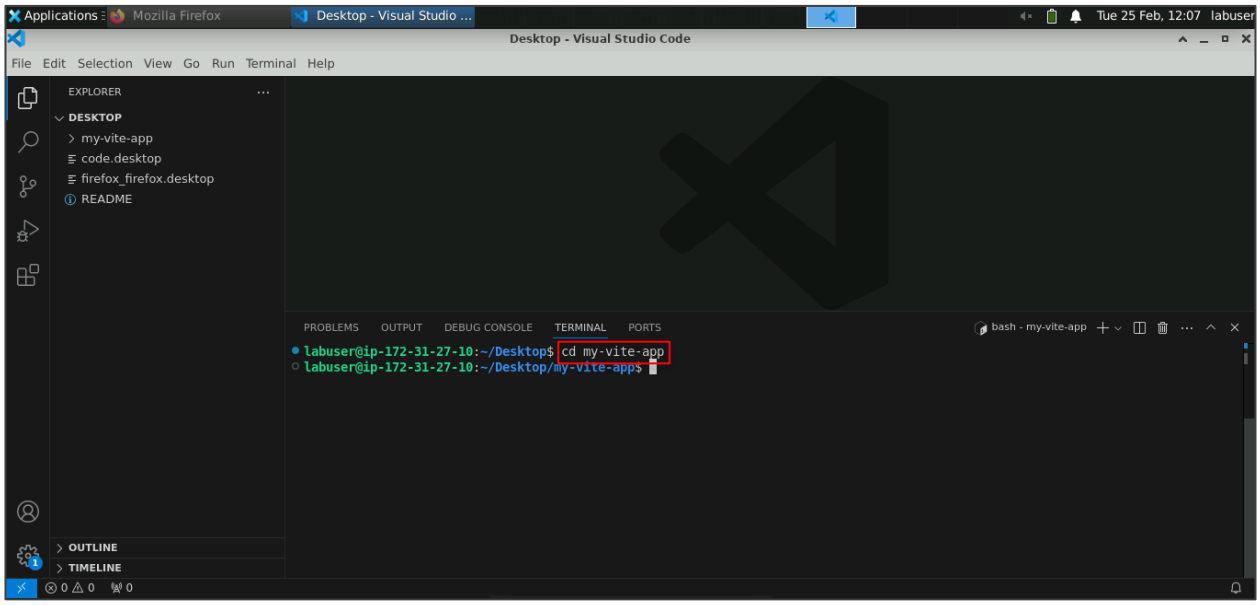


1.4 Select **JavaScript** as the language and press enter



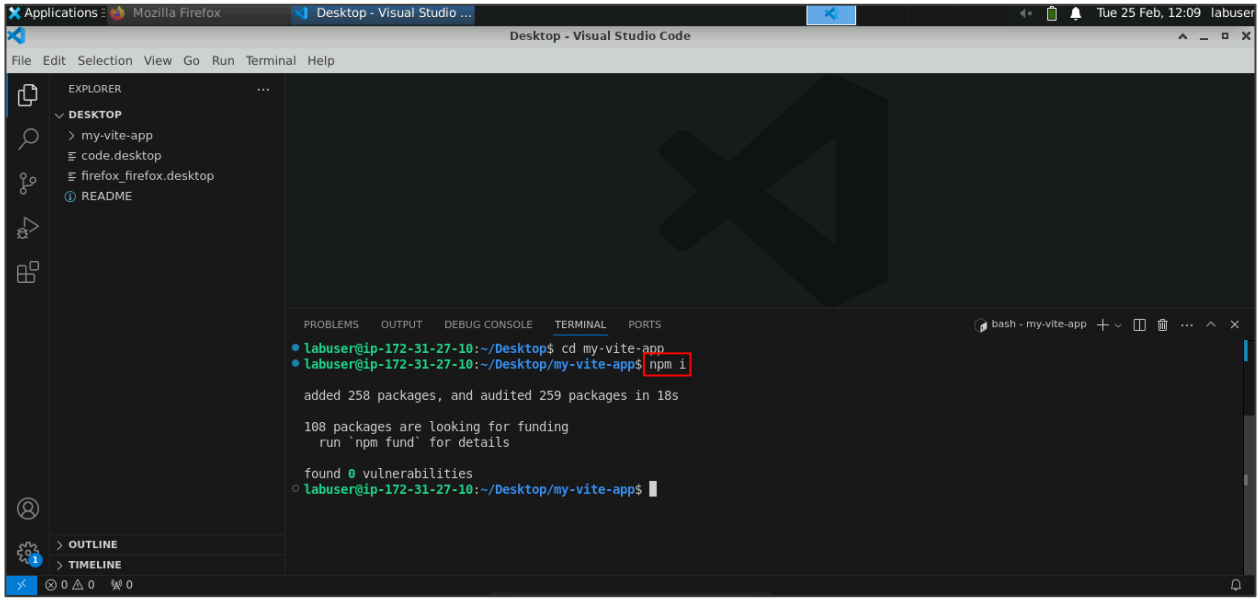
1.5 Run the following command to navigate to the project folder:

cd my-vite-app



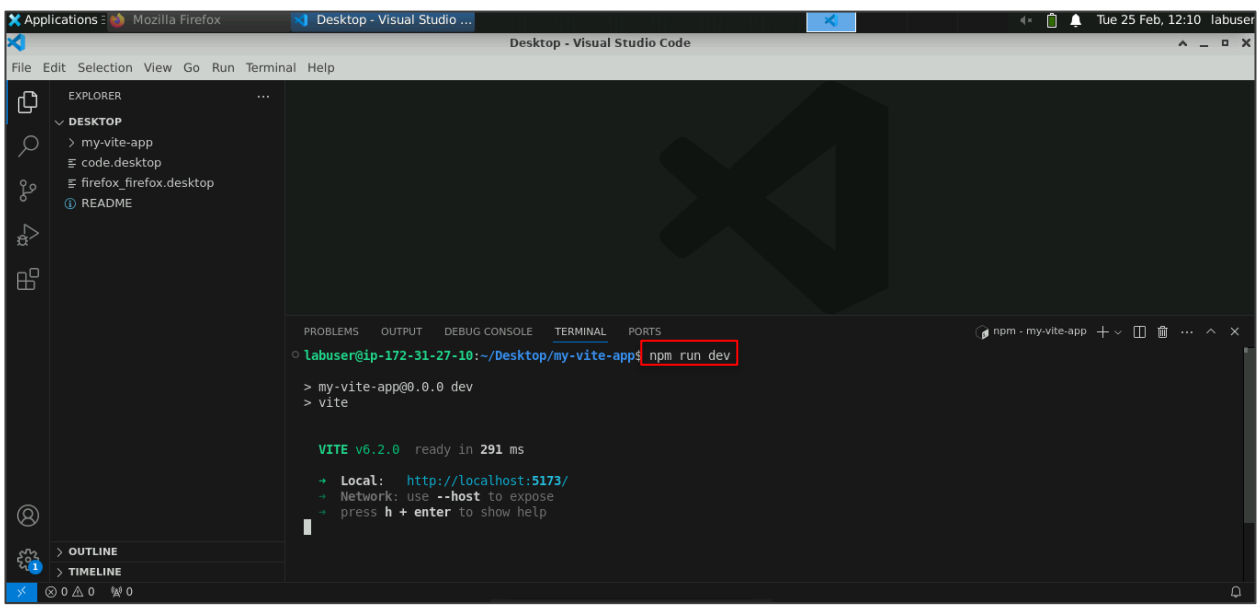
1.6 Run the following command to install dependencies:

npm i



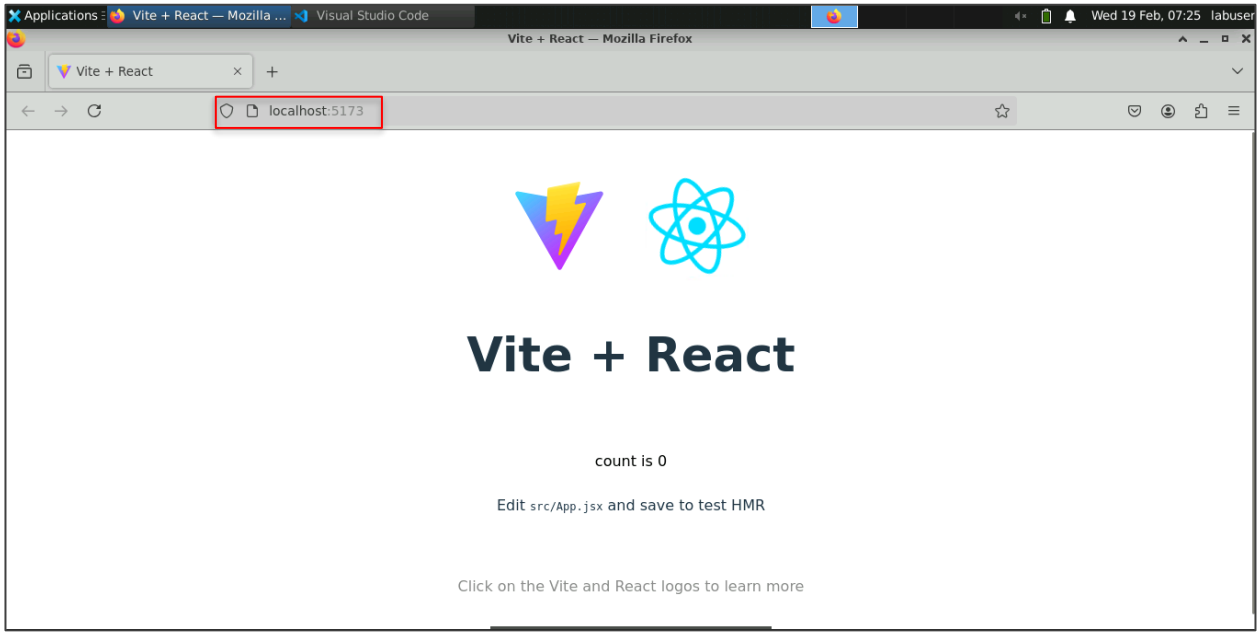
1.7 Run the following command to start the development server:

npm run dev



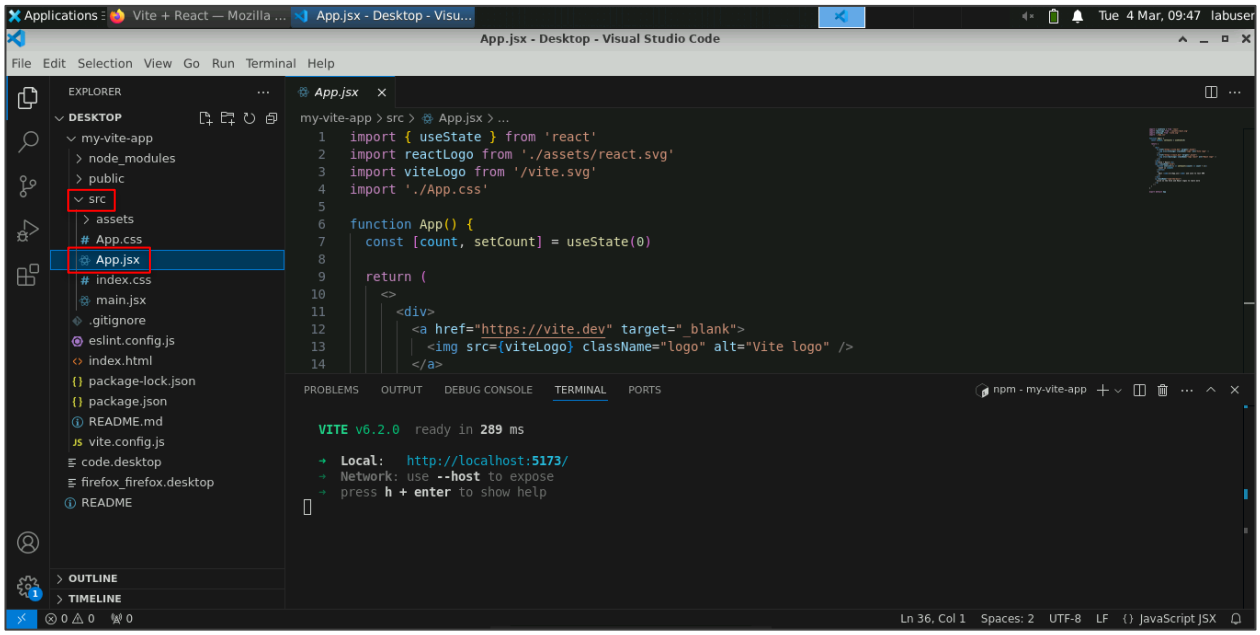
1.8 Navigate to the web browser and paste the link below:

http://localhost:5173



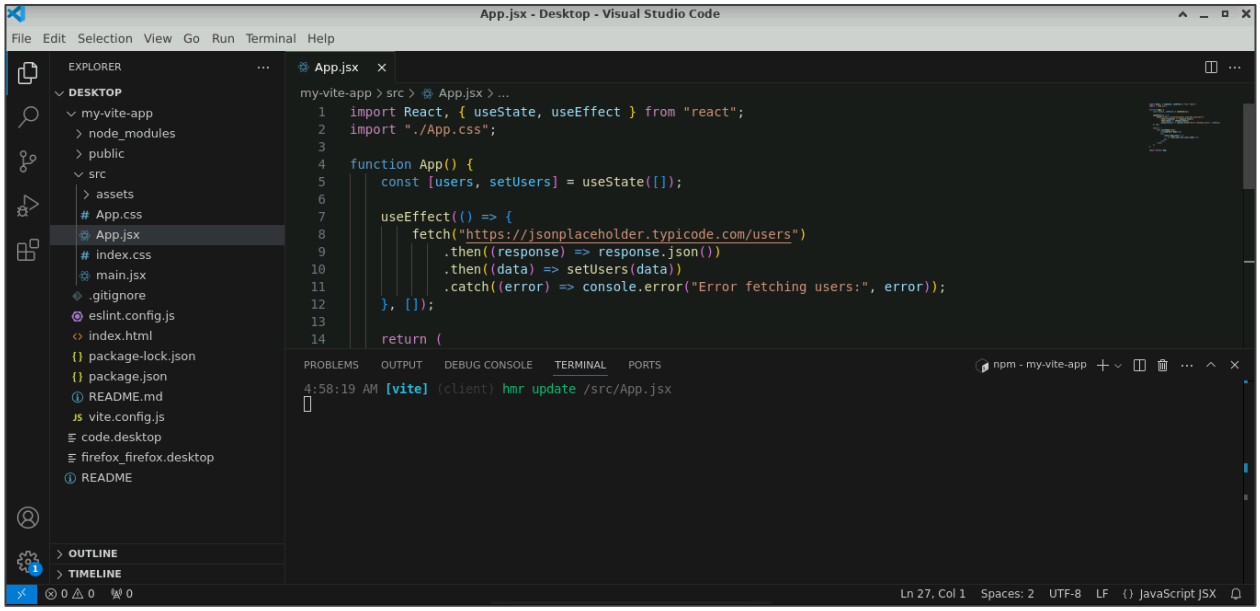
Step 2: Modify the App.jsx file

2.1 Open the **App.jsx** file inside the **src** folder



2.2 Paste the following code to include the new components in **App.jsx**:

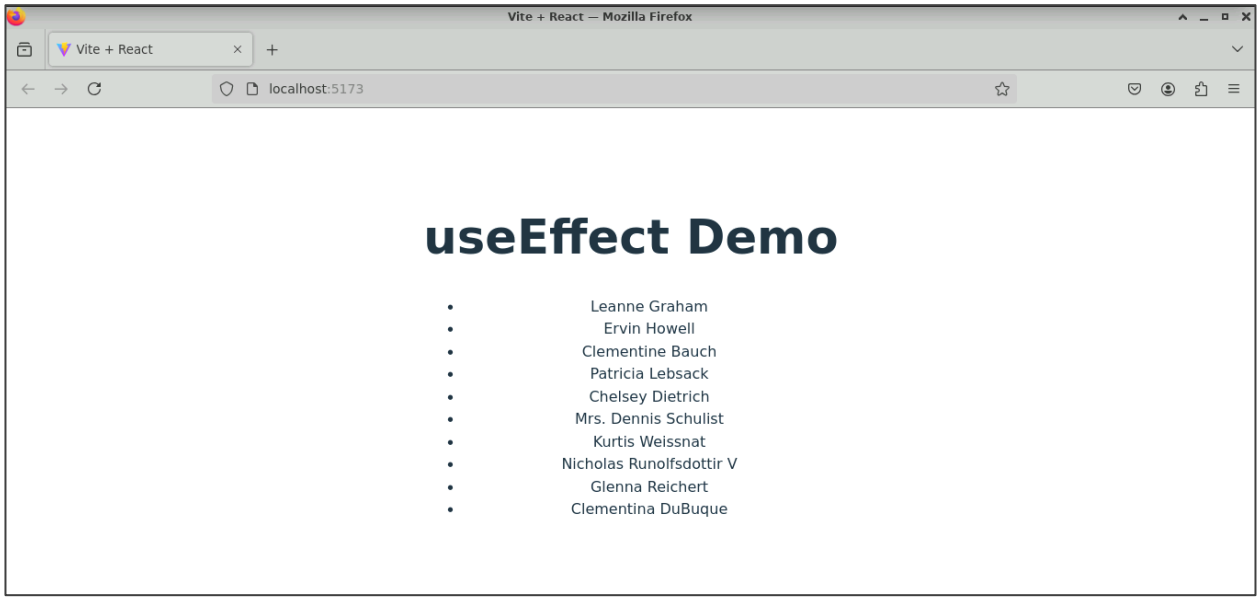
```
import React, { useState, useEffect } from "react";
import "./App.css";

function App() {
  const [users, setUsers] = useState([]);

  useEffect(() => {
    fetch("https://jsonplaceholder.typicode.com/users")
      .then((response) => response.json())
      .then((data) => setUsers(data))
      .catch((error) => console.error("Error fetching users:", error));
  }, []);

  return (
    <div className="App">
      <h1>useEffect Demo</h1>
      <ul>
        {users.map((user) => (
          <li key={user.id}>{user.name}</li>
        ))}
      </ul>
    </div>
  );
}

export default App;
```



Note: The fetch function sends a request to **https://jsonplaceholder.typicode.com/users** and returns a Promise that resolves to give the response. The response is then parsed into JSON format.

Step 3: Run and verify the application

3.1 Navigate to the browser and refresh the webpage



Note: The application should display a list of usernames retrieved from the **JSONPlaceholder API**.

By following these steps, you have successfully created a React application using Vite that demonstrates the useEffect Hook. The application dynamically fetches user data when the component mounts and efficiently updates the state for display.